

$$\min_Z \mathcal{IB} = \min_Z [-\mathcal{I}(Z; Y) + \beta \mathcal{I}(Z; X)] = \min_Z [(1 - \beta) \mathcal{H}(Y \mid Z) + \beta \mathcal{H}(Z \mid Y)]$$

$$\Delta_{\mathcal{ST}} = \mathbb{E}_{\mathcal{S}} [\|f(\boldsymbol{z}) - y\|_2] - \mathbb{E}_{\mathcal{T}} [\|f(\boldsymbol{z}) - y\|_2] \leq 2KQ \left(\mathcal{H}(Z \mid Y) \right)$$

$$\mathcal{H}(Z \mid Y) \leq \mathbb{E}_{y_i \sim \mathcal{Y}} \left[-\log(\epsilon) \text{IntDim}(\mathcal{M}_i) + \log \frac{K'}{C(\epsilon)} \right]$$